Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Surface Temperature Thematic Assembly Centre

Level 3S ODYSSEA Sea Surface Temperature Near Real Time product over the European North-West Shelf/Iberia Biscay Irish Sea area

SST_ATL_PHY_L3S_NRT_OBSERVATIONS_010_037

Issue: 2.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	16 June 2022	All	Initial release	Jean-François Piollé	A. Pisano
1.1	24 June 2022	1	updated statistics in version 1.0	Jean-François Piollé	A. Pisano
1.2	08 June 2023	All	Update for EIS Nov 2023 added Metop-C; addition of per sensor type multisensor grids.	Jean-François Piollé	A. Pisano
2.0	28/05/2025		Update for EIS Nov 2025 added NOAA-21 VIIRS as input sensor (table 3)	Jean-François Piollé	A. Pisano

ACRONYM TABLE

Acronym	Description
Argo	International profiling float network (www.argo.net)
EAN	Estimated Accuracy Number
GHRSST	Group for High Resolution Sea Surface Temperature
L2P	Level 2 Pre-processed
L4	Level 4 (Analysed)
NRT	Near Real Time
MY	Multi-year
NRT	Near Real Time
OSI SAF	Eumetsat Ocean and Sea Ice Satellite Application Facility
REP	Reprocessing
SST	Sea Surface Temperature
TAC	Copernicus Marine Service Thematic Assembly Centre

KEY DEFINITIONS

sea surface foundation temperature	**CF Definition**: The surface called "surface" means the lower boundary of the atmosphere. The sea surface foundation temperature is the water temperature that is not influenced by a thermally stratified layer of diurnal temperature variability (either by daytime warming or nocturnal cooling). The foundation temperature is named to indicate that it is the temperature from which the growth of the diurnal thermocline develops each day, noting that on some occasions with a deep mixed layer there is no clear foundation temperature in the surface layer. In general, sea surface foundation temperature will be similar to a night-time minimum or pre-dawn value at depths of between approximately 1 and 5 meters. In the absence of any diurnal signal, the foundation temperature is considered equivalent to the quantity with standard name sea_surface_subskin_temperature. The sea surface foundation temperature defines a level in the upper water column that varies in depth, space, and time depending on the local balance between thermal stratification and turbulent energy and is expected to change slowly over the course of a day. If possible, a data variable with the standard name sea_surface_foundation_temperature should be used with a scalar vertical coordinate variable to specify the depth of the foundation level. Sea surface foundation temperature is measured at the base of the diurnal thermocline or as close to the water surface as possible in the absence of thermal stratification. Only in situ contact thermometry is able to measure the sea surface foundation temperature. Analysis procedures must be used to estimate sea surface foundation temperature value from radiometric satellite measurements of the quantities with standard names sea_surface_skin_temperature and sea_surface_subskin_temperature. Sea surface foundation temperature provides a connection with the historical concept of a "bulk" sea surface temperature considered representative of the oceanic mixed layer temperature that is typically represented by any sea temperature measurement within the upper ocean over a depth range of 1 to approximately 20 meters. The general term, "bulk" sea surface temperature, has the standard name sea_surface_temperature with no associated vertical coordinate axis. Sea surface foundation temperature provides a more precise, well-defined quantity than "bulk" sea surface temperature and, consequently, is more representative of the mixed layer temperature. The temperature of sea water at a particular depth (other than the foundation level) should be reported using the standard name sea_water_temperature and, wherever possible, supplying a vertical coordinate axis or scalar coordinate variable.

RELEVANT DOCUMENT LIST

GHRSST GDS	The Recommended GHRSST Data Specification (GDS) Revision 2.1, June 2022, https://ghrsst.org
PUM	Related Product User Manual https://documentation.marine.copernicus.eu/PUM/CMEMS-SST-PUM-010-037.pdf

TABLE OF CONTENTS

Acronym Table *3*

Key definitions *4*

Relevant document List *5*

I. Executive summary *7*

I.1 Products covered by this document 7

I.2 Summary of the results 7

I.3 Estimated Accuracy Numbers 7

II. Production system description *8*

II.1 Production centre details 8

II.2 Short description of SST ODYSSEA Processing Chain and Algorithm 9

II.3 Collection of inputs 11

II.4 Editing and compositing 12

II.5 Bias correction 13

II.6 Merging 13

III Validation framework *15*

IV Validation results *16*

V System's Noticeable events, outages or changes *17*

VI Quality changes since previous version *18*

References *19*

I. EXECUTIVE SUMMARY

I.1 Products covered by this document

The Copernicus Marine Service SST-TAC (Sea Surface Temperature - Thematic Assembly Centre) SST product covered in this document is the SST_ATL_PHY_L3S_NRT_OBSERVATIONS_010_037, a multi-sensor level 3S (L3S) composite of SST observations, covering the European North West shelf/Iberia Biscay Irish Sea/Canary area at 0.02° horizontal resolution, produced by the Ifremer (Brest, France) production unit. This dataset results from the merging of various satellite SST level 2 (L2) data, which have passed a significant number of quality checks and editing, and which have been inter-calibrated through an inter-sensor bias correction procedure, using a median field generated from a set of "best quality" sensors, to provide an estimate of the night time SST based on original SST observations.

I.2 Summary of the results

The baseline for the validation of SST TAC products is to use drifting buoy measurements only, as recommended currently by the GHRSST[1] (Group for High Resolution Sea Surface temperature) group on satellite SST validation. The mean and standard deviation of departures from drifting buoy SST measurements are computed over the North West Shelf/Iberia Biscay Irish Sea/Canary area. Note that the validation results cover a one-year period, from 1st February 2021 to 31st December 2021. Integrated over the whole domain, the bias in surface temperature is -0.057°C with a standard deviation of 0.321°C.

I.3 Estimated Accuracy Numbers

The bias and standard deviation of the estimated sea surface temperature compared to buoys are provided in Table 1. These metrics are estimated over the whole domain over the time period from 1st February 2021 to 31st December 2021.

Table 1. Estimated product accuracy. Metrics: mean, standard deviation and number of match-ups for differences between High Resolution SST analysis over the whole ATL area and sub-regions European North West Shelf and Iberia Biscay Irish Sea (IBI ROOS) area and in-situ SST measurements provided by the INS TAC. The statistics were computed for the dates between 1st February 2021 and 31st December 2021.

Area	Mean bias	RMSD	Number of Match-ups
ATL (8°N to 62°N, 21°W to 13°E)	-0.056 ± 0.001	0.364 ± 0.001	74273
North Western Shelves (48°N to 62°N, 18°W to 13°E)	-0.056 ± 0.001	0.270 ± 0.001	11735
IBI ROOS (28°N to 56°N, 19°W to 5°E)	-0.062 ± 0.001	0.344 ± 0.001	47729

[1] https://www.ghrsst.org

II. PRODUCTION SYSTEM DESCRIPTION

II.1 Production centre details

The main properties of this dataset are provided in Table 2 below.

Table2. Details of the product

Production centre name	**IFREMER, FR**
Product name	SST_ATL_PHY_L3S_NRT_OBSERVATIONS_010_037
Short description	L3S daily maps of SST over North East Atlantic
Variables/datasets	adjusted_sea_surface_temperature
Reprocessing	No
Geographical coverage	North East Atlantic (ATL): North-West Shelf, Iberia, Biscay Irish Sea, Canary Islands: 9°N - 62°N, 21°W - 13°E
Spatial resolution	0.02x0.02 degree (equirectangular grid)
Vertical coverage	Surface
Temporal coverage	2021-01-01 – present
Temporal resolution	Daily
Target delivery time	Daily at 12pm (noon) UTC for day-1 00:00 UTC Analysis
Update Frequency	Daily

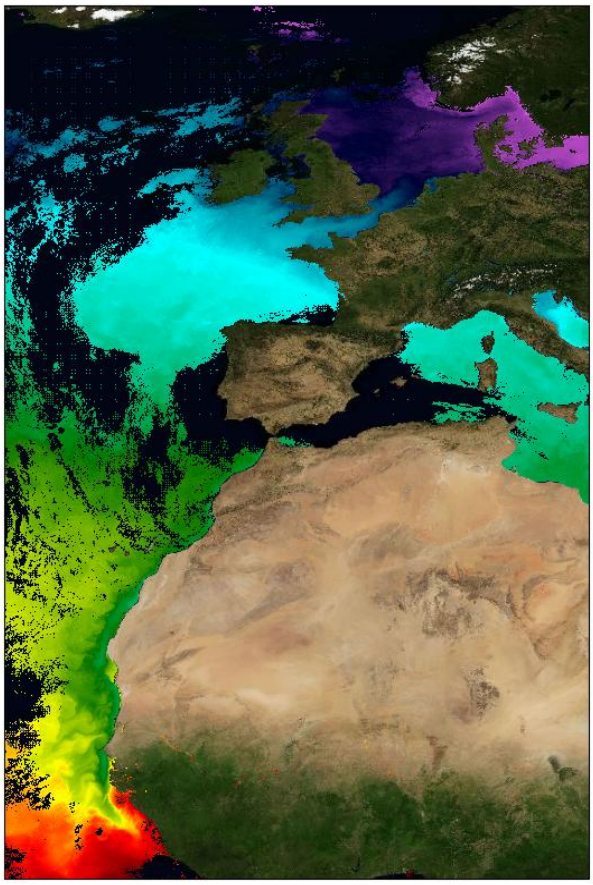

Figure 1: ATL area covered by this product.

II.2 Short description of SST ODYSSEA Processing Chain and Algorithm

The European North West Shelf/Iberia Biscay Irish Sea/Canary area (ATL) (extent shown in Figure 1).High Resolution ODYSSEA L3S Sea Surface Temperatures are produced with the Ifremer's ODYSSEA Sea Surface Temperature Level 4 (L4) processing chain, currently using (as of June 2025) the version 3.0.

Figure 2 shows the different steps for the processing of ODYSSEA SST products at Ifremer. The different steps are detailed hereafter. This dataset is the product generated at the merging step right before the optimal interpolation step that produces the ATL L4 product (SST_ATL_SST_L4_NRT_OBSERVATIONS_010_025), also available in the CMEMS.

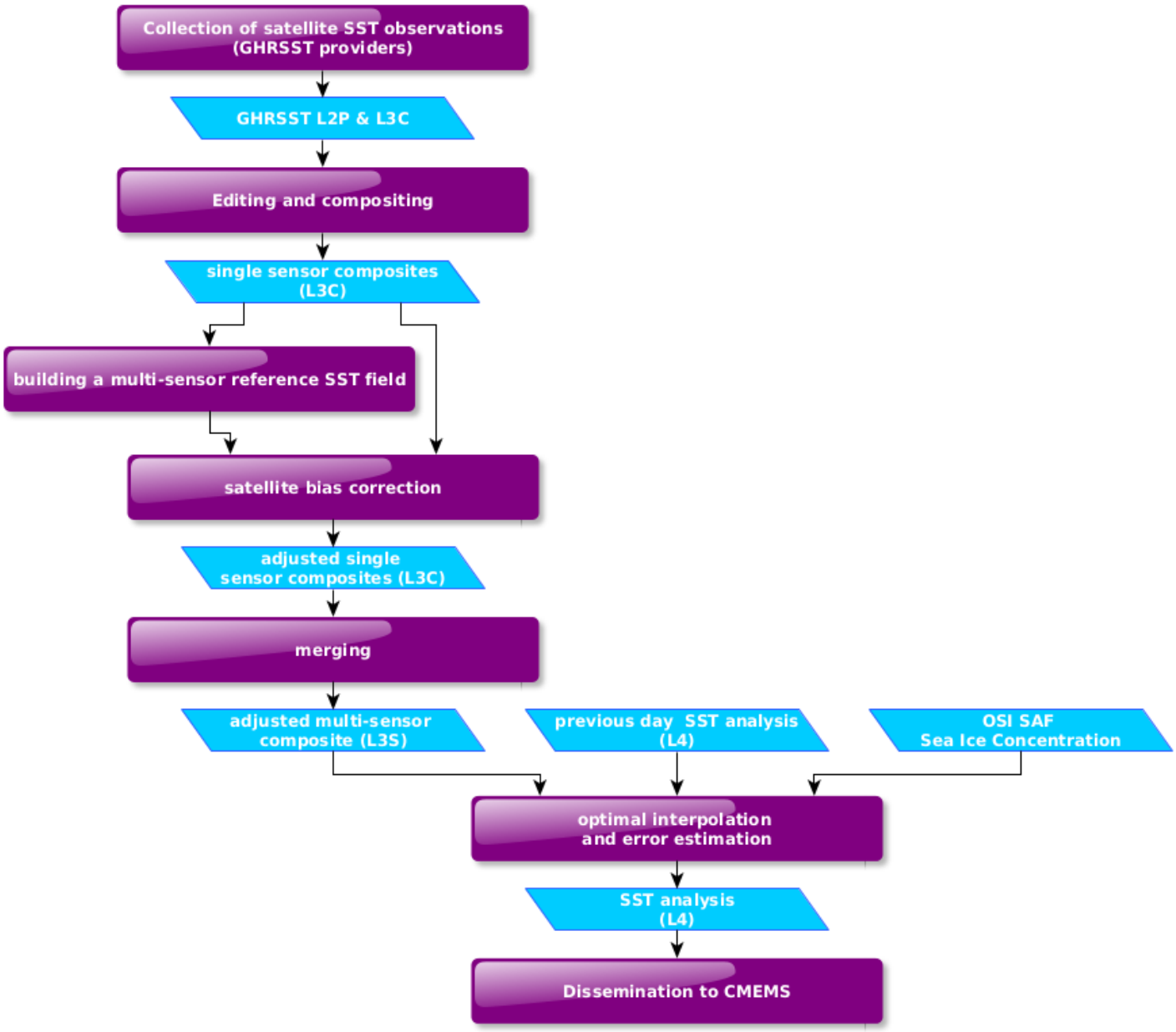

Figure 2: Schematic diagram of the ODYSSEA operational processing chain at Ifremer (note: OSI SAF: Ocean and Sea-ice Satellite Application Facility).

II.3 Collection of inputs

In the current configuration of the chain, a selection of GHRSST L2P (for polar orbiting satellites) and L3C (for geostationary satellites) products from different producers NASA, NOAA, EUMETSAT OSI-SA (Satellite Application Facility on Ocean and Sea Ice[2]), ESA) - in GHRSST compliant format [R.1] - are collected, to be merged into the final product.

All L2P/L3C data are provided with Sensor Specific Error Statistics (Single Sensor Error Statistic, SSES_bias and SSES_std variables) that give respectively an estimate of the systematic and random errors at pixel level. In addition, quality level flags are provided.

The L2P data collected from currently active missions or for older dates for ODYSSEA are listed in Table 3.

Table 3. Satellite L2P data collected for ODYSSEA (highlighted in grey the L2P currently ingested as of June 2025, others were used in previous years only).

Product	**Provider**	**Resolution**	**Used in intercalibration reference**	**Status**
AVHRR GAC NOAA17, 18,19	NAVO	9 km	Yes	Missions terminated
AVHRR LAC NOAA17,18,19	NAVO	1 km	Yes	Missions terminated
AATSR ENVISAT	ESA	1 km	Yes	Mission terminated
AVHRR METOP-B	OSI SAF	1 km	Yes	Active
AVHRR METOP-C	OSI SAF	1 km	Yes	Active
SEVIRI MSG	OSI SAF	5 km / 3hr	Yes	Active
AVHRR NAR NOAA17/18/19	OSI SAF	2 km	Yes	Missions terminated
AVHRR NAR /METOP-A	OSI SAF	1 km	Yes	Mission terminated
MODIS / AQUA	PODAAC	1 km	No	Replaced with other sources
NPP (Suomi) / VIIRS	OSPO	1 km	Yes	Active
NOAA-20 / VIIRS	OSPO	1 km	Yes	Active
NOAA-21 / VIIRS	OSPO	1 km	Yes	Active
AMSR-E	REMSS	30 km	Yes	Mission terminated
AMSR-2	REMSS	30 km	No	Active
TMI	REMSS	30 km	Yes	Mission terminated
Sentinel-3 A / SLSTR	EUMETSAT	1 km	Yes	Active
Sentinel-3 B / SLSTR	EUMETSAT	1 km	Yes	Active

[2] https://osi-saf.eumetsat.int/

The main sources for these data are:

- OSI SAF: EUMETSAT / Ocean and Sea-ice Satellite Application Facility
- NAVO: US Navy / Naval Oceanographic Office
- OSPO: NOAA (National Oceanic and Atmospheric Administration) / Office of Satellite and Product Operations
- PODAAC : NASA Physical Oceanography Distributed Active Archive Center
- REMSS : Remote Sensing Systems

In addition, the following complementary dataset are used:

- **Sea-ice concentration:** data from the EUMETSAT OSI-SAF[3] are used in ODYSSEA to mask out sea ice covered areas (though the area covered by this product is ice-free).
- **Climatology**: the SST climatology used has been derived from the PATHFINDER SST 5-daily climatology produced by [R.2]. The fields have been spatially smoothed to reduce possible artefacts, due to small scales and interpolated in time (daily). They are only used to spin up the L4 analysis and are not used in the L3S.

A land mask has been defined on the ODYSSEA v1.0 analysis grid. This land mask was extracted from GMT [R.3].

II.4 Editing and compositing

All satellite sea surface temperature (SST) data valid for a particular day (from previous day 12:00 to current day 12:00) are extracted from the input L2P/L3C files downloaded from each provider and re-gridded independently for each satellite mission onto the target grid to produce single sensor composite files. The input SST data undergo various editing and processing steps:

- only satellite data which have a quality flag above a certain level (specifically defined internally for the ODYSSEA system for each satellite) and higher are accepted. This minimum quality level, a variable available in all GHRSST input datasets [R.1] can be different depending on the data source and is adjusted per source after expert evaluation input from single sensor comparisons with in situ drifter measurements.
- day-time data, which are determined using a solar zenith angle calculation are rejected and only nighttime data are kept to provide SST close to the foundation (free of diurnal cycle variations) sea surface temperature value.
- the SSES biases supplied with the GHRSST data are removed from each pixel and the SSES standard deviation values are passed on to the next steps in the analysis chain.
- for skin temperature datasets (e.g. SLSTR on S3A and S3B), a skin-to-subskin correction is applied [R.2]. Measurements where wind speed is lower than 2 m/s are ignored.
- incidence angle for geostationary sensors is also taken into account to remove pixels with too high an incidence angle (greater than 55°).
- ice concentration, when provided in the L2P/L3C (at high latitudes) an ice mask from OSI SAF is used to remove pixels with ice contamination greater than 10%.

[3] https://osi-saf.eumetsat.int/

- aerosols, when provided in the L2P /L3C is used to remove pixels with aerosols content greater than 0.3.
- various filters may be applied too, depending on the source (valid range for SST, type of SST retrieval algorithm, etc.).
- pixels having the highest quality and falling into the same target grid cells are averaged together, if a minimum number of clear-sky pixels varying with respect to the sensor and final grid resolution is available, in order to avoid unrepresentative grid cells.

II.5 Bias correction

Satellite data can be biased for several reasons, including: atmospheric water vapour; atmospheric aerosol (dust); surface changes (e.g. extreme roughness); and instrument calibration problems. These effects can lead to biases with respect to in situ measurements and spatio-temporal discrepancies in the analysis, if they are not addressed in some way. In order to prepare a consistent daily dataset, an intercalibration method has been developed. This processing step consists of estimating a daily correction to be applied on SST values for each single satellite source. The main issue is to define a daily reference to estimate this correction. Then for each single source, a large-scale bias field (with synoptic weather scales) is estimated from the differences between the observations and the reference of the day, followed by the application of the estimated correction on SST values for each satellite source.

The current methodology (internal) implemented as of June 2025 consists of building a reference SST field (on a 0.10-degree grid), combining the measurements of the “best source” (currently VIIRS onboard NPP as processed by NOAA/OSPO, as selected in ODYSSEA by comparisons with in situ drifter measurements) with the median value of a selection of trusted sources (refer to Table 3 for the list of products currently taken into account in this median calculation). The choice of the best source and the trusted sources for the median calculation was evaluated from comparisons with in situ measurements over one year of input data.

The difference between each source to be merged into the final product and the computed reference is then smoothed with a large scale gaussian filters and removed from the original SST to provide an **adjusted sea surface temperature**.

The outcome of this process is a new version of the single composite satellite data (L3C), which have been bias-corrected (or adjusted).

II.6 Merging

The daily single sensor composite files are then merged together into a combined multi-sensor composite file (L3S) at the final product resolution. The single sensor composites are first re-gridded, if necessary, on the final grid resolution using nearest neighbour or averaging, depending on whether the single sensor composites are of coarser or higher resolution than the final grid. The merging process then takes into account a hierarchy of sensors (from best to worst), estimated again by comparing each source with in situ measurements. The polar orbiting infrared sensors, polar orbiting microwave sensors (AMSR2) and geostationary infrared sensors are merged together onto separate grids (respectively 0.02-degree, 0.25 degree and 0.05 degree resolution) and then onto a final 0.02-degree resolution grid (with priority order: polar orbiting infrared, geostationary infrared, polar orbiting microwave sensor). The

final merge and the per sensor-type grids are all provided in separate grids. Users who want to account for the true resolution of the sensor observation should use the per sensor-type grids.

A given cell of the final grid will contain a value from one source only (the best selected). The used source is reported in the *sources_of_sst* variable of the created L3S.

III VALIDATION FRAMEWORK

The baseline for the validation of SST TAC products is to use drifting buoy measurements only, as recommended currently by the GHRSST group on satellite SST validation (STVAL). The mean and standard deviation difference against drifting buoy SST measurements are computed over the North West Shelf/Iberia Biscay Irish Sea/Canary area. Various quality check procedures are applied on drifting buoy measurements to discard suspect data, using gross error checks and blacklists (identifiers of bad buoys) provided by UK MetOffice and Meteo-France/CMS identifying buoys reporting wrong measurements. Only night-time measurements are kept, to be consistent with the data screening policy for the input satellite data.

Overall the L3S SST product over European Seas is purely based on satellite observations, and hence drifting buoy measurements can be considered to be a fully independent validation data. However, in some regional seas, the number of drifting buoy measurements is very low (e.g. Baltic Sea), and even equal to zero. In the future, other data sources (moored buoys, ARGO floats, etc.) could be considered for validation statistics if available.

The present validation study specifically uses drifting buoy measurements provided by the CMEMS IN-SITU TAC. The analysis minus the observation differences are calculated by averaging the in situ data over the same grid cell as the satellite dataset. The mean bias and RMSD based on these differences are then computed.

These diagnostics are processed for every day of the time series and over the validation year, for the full domain and for the different sub-regions.

IV VALIDATION RESULTS

Table 4 shows the mean bias, standard deviation and number of observations of the ATL L3S observations minus drifting buoy differences for the whole ATL domain, for the whole ATL area and the two validation sub-regions (North Western Shelves and IBI ROOS) for the period 01/02/2021 to 31/12/2021.

Table 4. Mean, RMSD, and number of match-ups for differences between High Resolution SST analysis over the European North West Shelf/Iberia Biscay Irish Sea/Canary area (ATL) and the two sub regions (North Western Shelves and IBI ROOS) and in-situ SST measurements provided by the INS TAC. The statistics were computed for the dates between 1st February 2021 and 31st December 2021.

Area	Mean bias (°C)	RMSD (°C)	Number of Match-ups
ATL (8°N to 62°N, 21°W to 13°E)	-0.056 ± 0.001	0.364 ± 0.001	74273
North Western Shelves (48°N to 62°N, 18°W to 13°E)	-0.056 ± 0.001	0.270 ± 0.001	11735
IBI ROOS (28°N to 56°N, 19°W to 5°E)	-0.062 ± 0.001	0.344 ± 0.001	47729

A visual distribution of the statistical metrics is shown in Figure 3 for the same time period (1st February 2022 to 31st December 2021). This shows the variation of the mean bias and number of match-ups over the area in the validation period.

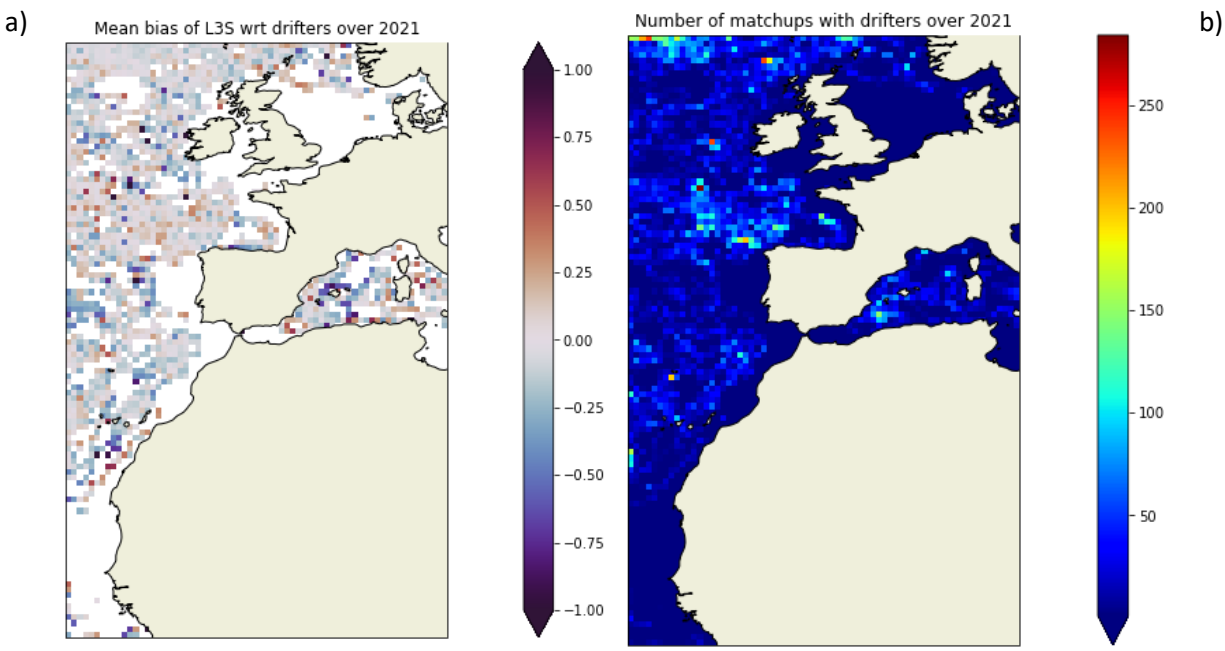

Figure 3: a) Spatial distribution of mean bias (°C), and b) number of available match-ups for differences computed over the period from 1st February 2021 and 31st December 2021 (white areas in mean bias correspond to no data).

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality (Table 5).

Table 5. System's noticeable changes.

Date	Change/Event description	System version	Other
November 2022	Initial version	3.0	
November 2023	Changed L3S format (multi-resolution grids, per sensor type) and added Metop-C in input data streams	3.0	
November 2025	Addition of NOAA-21/VIIRS as a new input data stream	3.0	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

The previous version was released in November 2023. This update covers the addition of NOAA-21/VIIRS in the list of input data streams to this multi mission dataset, to anticipate the decommissioning of NOAA-NPP/VIIRS. The sensor and processing for these two inputs are identical (and done by NOAA).

The quality of the dataset remains in the same range as the previous release. The product was generated over a period of 3 months (Oct to Dec 2024) with and without NOAA-21 VIIRS : with a mean difference of -0.008 K and RMSD of 0.09 K, we concluded the addition of NOAA-21 does not impact the product quality and the validation results remain consistent (comparisons with in situ showed similar results). The addition of this new source increases on average by about 8% the amount of observations in the dataset which is significant in terms of additional coverage but only temporary as we have to account for the upcoming decommissioning of NOAA-NPP (expected 2028): it still improves the robustness of the dataset production as NOAA-21 provides backup in case one or more of the other mission input datasets becomes unavailable (e.g., satellite manoeuvre, sensor malfunction, data provision issue, etc.).

REFERENCES

[R.1] The GHRSST Data Processing Specification (GDS) provides a detailed technical specification for producing GHRSST L2, L3 and L4 products. The latest version of this document is the GDS 2.1 version 0, published in summer 2022. The Group for High Resolution Sea Surface Temperature Science Team, Piollé, Jean-François, Armstrong, Ed, Casey, Kenneth, & Donlon, Craig. (2022). The Recommended GHRSST Data Specification (GDS) (GDS 2.1 revision 0). Zenodo. https://doi.org/10.5281/zenodo.6984989

[R.2] Wessel, P., and W. H. F. Smith, New, improved version of Generic Mapping Tools released, EOS Trans.Amer. Geophys. U., vol. 79 (47), pp. 579, 1998

[R.3] Donlon, C. J., P. J. Minnett, C. Gentemann, T. J. Nightingale, I. J. Barton, B. Ward, and M. J. Murray, 2002: Toward Improved Validation of Satellite Sea Surface Skin Temperature Measurements for Climate Research. J. Climate, 15, 353–369, https://doi.org/10.1175/1520-0442(2002)015<0353:TIVOSS>2.0.CO;2